

SATHVIK S

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EDUCATION

PES University

Bangalore, India

Bachelor of Technology in Computer Science Engineering

- Current CGPA: 6.2

Alva's PU College

Moodbidri, D.K.

- Class 12th Percentage: 92.3%

EXPERIENCE

PESU ISFCR 5G Labs

Research Intern

- Assisted in deploying and configuring a 5G network using Linux systems and Open5GS in a lab environment
- Troubleshoot and resolved configuration issues to ensure seamless integration and operation of 5G network components
- Assisted in initial configurations related to 5G network slicing within the Open5GS environment for service differentiation

RESEARCH / PROJECTS

EmoStream: Concurrent Emoji Broadcast over Event-Driven Architecture

Python | Spark | Kafka | Flask | Redis

- Designed and implemented a scalable system to process millions of real-time emoji reactions during live sporting events using Kafka and Spark Streaming.
- Developed an asynchronous Flask API endpoint to handle high-concurrency input from multiple clients per second and stream data to Kafka brokers.
- Built a Spark Streaming job for 2-second micro-batch processing, aggregating similar emoji reactions into single representative counts.
- Established a Pub-Sub architecture with cluster publishers and subscribers to ensure real-time delivery of aggregated data to thousands of users.

Competitive Exam Rank Automation Platform

Python | MySQL | YAML

- Built a Python-based system to generate and manage JEE Mains student data using YAML scripts and MySQL, including registration IDs, scores, and ranks
- Designed SQL views for real-time percentile and rank calculations with customizable tie-breaker logic across multiple sessions
- Automated aggregation and analysis of nationwide JEE Mains data, handling duplicates and tracking multiple attempts efficiently

Hybrid Watermarking Framework with AI-Powered Recovery and Blockchain Ownership Verification

Python | Deep Learning | OpenCV | Blockchain

- Designed a secure digital image watermarking framework integrating AI-based recovery mechanisms for enhanced asset protection.
- Implemented blockchain technology to verify ownership, ensuring the immutability and traceability of the watermarked images.
- Applied advanced digital image processing and deep learning models to strengthen the watermark's robustness against complex extraction attacks.

Analytic System for Predicting and Reducing Customer Churn

PowerBI | Scikit-learn | OpenCV | PyTorch | MongoDB | FastAPI | Chart.js/Recharts

- Analyzed data to identify key factors influencing churn (e.g., delivery delays, low engagement), while carefully separating correlation from real causal influence.
- Estimated the true impact of potential actions (discounts, faster delivery, etc.) on churn probability reduction using causal/uplift modelling.
- Compared actions based on business impact and recommended the action delivering the maximum net benefit.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, HTML/CSS, Rust, SQL

Tools: Docker, Kubernetes, Redis, MongoDB, PyTorch, Pandas, NumPy, Git, MySQL, YAML, Kafka, Spark